

Approximating the score function using Optimal Transport

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We propose a novel framework to approximate the gradient flow of the free energy functional in Wasserstein space via Kantorovich potential. By reformulating the Jordan-Kinderlehrer-Otto (JKO) scheme as a sequence of convex sub-problems, our approach leverages stochastic gradient descent (SGD) to efficiently solve each minimisation problem. We study the semi-discrete case, where one measure (the source one) is seen as a discrete over the samples. In this scenario, we establish the convergence of the SGD iterates and provide error estimates. Numerical experiments illustrate the potential of our method in learning complex data distributions.

Joint work with: Thomas Gallouët, Ziad Kobeissi.

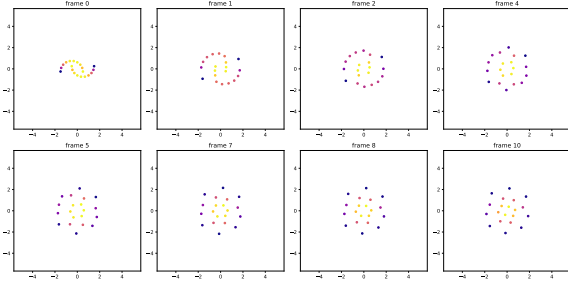


Figure 1: Forward: resulting points clouds after JKO steps 0, 1, 2, 4 (first row), 5, 7, 8 and 10 (second row). Simulations made with `sdot`.

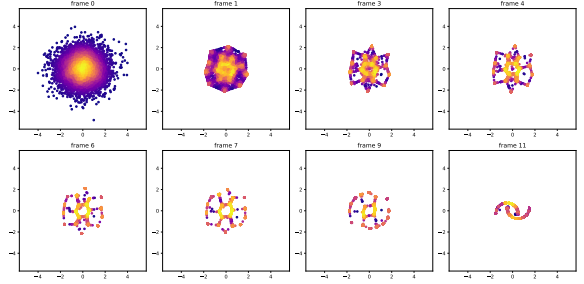


Figure 2: Backward: resulting points clouds after inverse steps 0, 1, 2, 4 (first row), 5, 7, 8 and 11 (second row). Simulations made with `sdot`.

References

- [1] R. Jordan, D. Kinderlehrer and F. Otto. The Variational Formulation of the Fokker–Planck Equation. *SIAM Journal on Mathematical Analysis*, 10.1137/S0036141096303359, 1998
- [2] F. Genans, A. Goddichon-Baggioni, F.-X. Vialard and O. Wintenberger. Stochastic Optimization in Semi-Discrete Optimal Transport: Convergence Analysis and Minimax Rate. *NeurIPS*, 10.48550/arXiv.2510.25287, 2025